Solution A: Amortized Inference of the Hadronic Shower Profile with Neptune by Conditional Negative Log-Likelihood Minimization

1 Inference problem

Definition 1 (Shower and target)**.** A simulated hadronic cascade is a random outcome ω (the full Geant4 history) with vertex $\boldsymbol{v} \in \mathbb{R}^3$, direction $\hat{\boldsymbol{u}} \in S^2$ and energy $E > 0$. Its charged-track-length profile along the axis is

$$\ell_\omega(s) = \sum_{\text{segments } i} w_i \, \delta(s - s_i), \qquad s_i = (\boldsymbol{r}_i - \boldsymbol{v}) \cdot \hat{\boldsymbol{u}}, \tag{1}$$

with w_i the segment lengths and $\boldsymbol{r}_i$ the segment midpoints. The target is the pair of log Gamma parameters $\boldsymbol{y} = T(\omega) = (\ln a_\omega, \ \ln b_\omega) \in \mathbb{R}^2$, where (a_ω, b_ω) maximize the length-weighted Gamma likelihood

$$(a_\omega, b_\omega) = \operatorname*{arg\,max}_{a,b>0} \ \sum_i w_i \Big[(a-1) \ln s_i - \frac{s_i}{b} - \ln \Gamma(a) - a \ln b \Big]. \tag{2}$$

Definition 2 (Observation)**.** The detector response is a finite set of hits

$$\mathcal{D} = \{(\boldsymbol{\xi}_j, \ \boldsymbol{f}_j)\}_{j=1}^n, \qquad \boldsymbol{\xi}_j = (x_j, y_j, z_j, t_j) \in \mathbb{R}^4, \quad \boldsymbol{f}_j \in \mathbb{R}^F, \tag{3}$$

one element per hit DOM, with $\boldsymbol{f}_j$ the F per-DOM summary features of its pulse series (charge, first-hit time, time quantiles, ...). n varies from event to event.

Data-generating process. Simulation draws $\omega \sim p_{\text{gen}}(\omega)$, then $\mathcal{D} \sim p(\mathcal{D} \mid \omega)$ (photon propagation and detector response). This induces the joint law of $(\boldsymbol{y}, \mathcal{D})$:

$$p_{\text{gen}}(\boldsymbol{y}, \mathcal{D}) = \int p(\mathcal{D} \mid \omega) \ \delta\big(\boldsymbol{y} - T(\omega)\big) \ p_{\text{gen}}(\omega) \, d\omega. \tag{4}$$

Goal. For an observed event $\mathcal{D}^*$, report the posterior

$$p(\boldsymbol{y} \mid \mathcal{D}^*) = \frac{p(\boldsymbol{y}, \mathcal{D}^*)}{\int p(\boldsymbol{y}', \mathcal{D}^*) \, d\boldsymbol{y}'}. \tag{5}$$

This density is intractable: it marginalizes over every shower history ω compatible with $\boldsymbol{y}$ and over the geometry. Solution A approximates it directly with a parametric conditional density $q_\phi(\boldsymbol{y} \mid \mathcal{D})$ that is *amortized*: one set of parameters ϕ serves every event.

2 Objective: conditional negative log-likelihood

Population objective. Choose ϕ to minimize the expected Kullback–Leibler divergence from the true posterior to the model:

$$\mathcal{R}(\phi) = \mathbb{E}_{p(\mathcal{D})}\Big[\mathrm{KL}\big(p(\boldsymbol{y} \mid \mathcal{D}) \,\big\|\, q_\phi(\boldsymbol{y} \mid \mathcal{D})\big)\Big]. \tag{6}$$

Expanding the KL divergence,

$$\mathcal{R}(\phi) = \underbrace{-\,\mathbb{E}_{p(\boldsymbol{y},\mathcal{D})}\big[\ln q_\phi(\boldsymbol{y} \mid \mathcal{D})\big]}_{\text{expected conditional NLL}} - \underbrace{H(\boldsymbol{Y} \mid \mathcal{D})}_{\text{independent of } \phi}, \tag{7}$$

so minimizing $\mathcal{R}$ is equivalent to minimizing the expected conditional negative log-likelihood. Only samples $(\boldsymbol{y}, \mathcal{D})$ are required, never the density $p(\boldsymbol{y} \mid \mathcal{D})$ itself. Here $\boldsymbol{Y}$ denotes the random target, and $H(\boldsymbol{Y} \mid \mathcal{D}) = -\mathbb{E}_{p(\boldsymbol{y},\mathcal{D})}[\ln p(\boldsymbol{y} \mid \mathcal{D})]$.

Target prior and importance weights. The simulated spectrum p_{gen} is generally not the prior one wants the posterior to reflect. For a target prior $\pi(\omega)$ with the same support, define the weights

$$w(\omega) = \frac{\pi(\omega)}{p_{\mathrm{gen}}(\omega)}, \qquad \text{e.g.} \quad w \propto E^{\gamma_{\mathrm{gen}} - \gamma_{\mathrm{target}}} \quad \text{for power-law spectra.} \tag{8}$$

Then $\mathbb{E}_\pi[f] = \mathbb{E}_{p_{\mathrm{gen}}}[wf]$ for any integrable f. The learned posterior is $p_\pi(\boldsymbol{y} \mid \mathcal{D})$, so the reweighting is part of the model and not only a variance-reduction device.

Empirical objective. With M independent simulated events $(\boldsymbol{y}_e, \mathcal{D}_e, w_e)$,

$$\boxed{\hat{\mathcal{J}}_M(\phi) = -\frac{\sum_{e=1}^{M} w_e \, \ln q_\phi(\boldsymbol{y}_e \mid \mathcal{D}_e)}{\sum_{e=1}^{M} w_e} + \frac{\lambda}{2}\|\phi\|_2^2} \tag{9}$$

The events are independent draws, so the log-likelihood of the training set is a (weighted) sum over events. This is the conditional independence assumption applied at the level of whole events. The L_2 term corresponds to a Gaussian prior on ϕ, making $\hat{\phi}$ a MAP estimate. AdamW's decoupled weight decay matches this only approximately.

Stochastic optimization. For a minibatch $\mathcal{B} \subset \{1, \ldots, M\}$ drawn uniformly,

$$\boldsymbol{g}_{\mathcal{B}} = -\frac{1}{\sum_{e \in \mathcal{B}} w_e} \sum_{e \in \mathcal{B}} w_e \, \nabla_\phi \ln q_\phi(\boldsymbol{y}_e \mid \mathcal{D}_e) + \lambda\phi \tag{10}$$

is a consistent (self-normalized) estimator of $\nabla_\phi \hat{\mathcal{J}}_M$.

3 Model: Neptune as a conditional density

The model factorizes as an encoder $\boldsymbol{c}_\phi : \mathcal{D} \mapsto \mathbb{R}^d$ followed by a density head, $q_\phi(\boldsymbol{y} \mid \mathcal{D}) = q_\phi\big(\boldsymbol{y} \mid \boldsymbol{c}_\phi(\mathcal{D})\big)$.

1. Point embedding. $\boldsymbol{z}_j = \mathrm{MLP}_1(\boldsymbol{f}_j) \in \mathbb{R}^{d_0}$.

2. Tokenization (farthest-point sampling). With the space–time metric $\|\boldsymbol{\xi}\|_\kappa^2 = x^2 + y^2 + z^2 + (\kappa t)^2$, where κ has units of velocity (default $0.3\,\mathrm{km}/\mu\mathrm{s}$), choose $T = \min(n, T_{\max})$ centroids greedily,

$$c_1 \text{ fixed}, \qquad c_{k+1} = \arg\max_j \ \min_{l \le k} \ \|\boldsymbol{\xi}_j - \boldsymbol{\xi}_{c_l}\|_\kappa, \tag{11}$$

and assign each hit to its nearest centroid (a Voronoi cell), $\alpha(j) = \arg\min_k \|\boldsymbol{\xi}_j - \boldsymbol{\xi}_{c_k}\|_\kappa$. Token k is

$$\boldsymbol{x}_k^{(0)} = \mathrm{MLP}_2\Big(\mathrm{pool}_{\{j:\alpha(j)=k\}} \boldsymbol{z}_j \ \oplus \ \big[\, n_k,\ Q_k,\ \mathrm{sd}(t)_k,\ r_k^{\mathrm{rms}} \,\big]\Big), \tag{12}$$

where n_k, Q_k, $\mathrm{sd}(t)_k$ and r_k^{rms} are the hit multiplicity, total charge, time spread and RMS radius of cell k, and the anchor of token k is $\boldsymbol{\xi}_k := \boldsymbol{\xi}_{c_k}$. If $n \le T_{\max}$, every hit is its own token.

3. Transformer encoder with 4-D rotary position encoding. Each of N_L layers applies, with RMS normalization $\mathrm{N}(\cdot)$,

$$X \leftarrow X + \mathrm{Attn}\big(\mathrm{N}(X)\big), \qquad X \leftarrow X + \mathrm{SwiGLU}\big(\mathrm{N}(X)\big). \tag{13}$$

For one head with queries $\boldsymbol{q}_k$, keys $\boldsymbol{k}_l$ and values $\boldsymbol{v}_l$, the rotary encoding applies a block-diagonal rotation $R(\boldsymbol{\xi})$ built from all four coordinates, with $R(\boldsymbol{\xi})^\top R(\boldsymbol{\xi}') = R(\boldsymbol{\xi}' - \boldsymbol{\xi})$:

$$A_{kl} = \frac{\big\langle R(\boldsymbol{\xi}_k)\boldsymbol{q}_k,\ R(\boldsymbol{\xi}_l)\boldsymbol{k}_l \big\rangle}{\sqrt{d_h}} = \frac{\boldsymbol{q}_k^\top R(\boldsymbol{\xi}_l - \boldsymbol{\xi}_k)\, \boldsymbol{k}_l}{\sqrt{d_h}}, \qquad \mathrm{Attn}(X)_k = \sum_{l \in \mathcal{V}} \mathrm{softmax}_l(A_{kl})\, \boldsymbol{v}_l, \tag{14}$$

where $\mathcal{V}$ is the set of valid (unpadded) tokens. The attention scores therefore depend on token positions only through the relative space–time offsets $\boldsymbol{\xi}_l - \boldsymbol{\xi}_k$. An additive Fourier encoding of absolute position is also used, so the model can still learn detector-specific effects such as depth-dependent ice.

4. Pooling and head. With attention weights $\beta_k = \mathrm{softmax}_{k \in \mathcal{V}}\big(\boldsymbol{a}^\top \boldsymbol{x}_k^{(N_L)}\big)$,

$$\boldsymbol{c}_\phi(\mathcal{D}) = \Big(\sum_{k \in \mathcal{V}} \beta_k\, \boldsymbol{x}_k^{(N_L)}\Big) \ \oplus \ \big[\ln(1+T),\ \ln(1+Q_{\mathrm{tot}})\big]. \tag{15}$$

The two scalars restore the event size and total light yield, which pooling normalizes away.

Remark 1 (Symmetry)**.** *Steps 1–4 act on the hit* set *and do not depend on the order of the hits, except through the choice of the first centroid c_1 in farthest-point sampling. q_ϕ is therefore permutation-invariant up to that tie-break.*

4 Density head

4.1 Gaussian head

The head outputs $\boldsymbol{\mu}_\phi(\boldsymbol{c}) \in \mathbb{R}^2$ and the entries of a lower-triangular matrix $L_\phi(\boldsymbol{c})$ with $(L_\phi)_{ii} = \mathrm{softplus}(\cdot) + \varepsilon > 0$. Set $\Sigma_\phi = L_\phi L_\phi^\top \succ 0$ and

$$q_\phi(\boldsymbol{y} \mid \mathcal{D}) = \mathcal{N}\big(\boldsymbol{y};\ \boldsymbol{\mu}_\phi,\ \Sigma_\phi\big). \tag{16}$$

The per-event loss is

$$-\ln q_\phi(\boldsymbol{y} \mid \mathcal{D}) = \tfrac{1}{2}\left\|L_\phi^{-1}(\boldsymbol{y}-\boldsymbol{\mu}_\phi)\right\|_2^2 + \sum_{i=1}^{2}\ln(L_\phi)_{ii} + \ln(2\pi), \tag{17}$$

using $\frac{1}{2}\ln\det\Sigma = \sum_i \ln L_{ii}$. The residual term costs one triangular solve. With L_ϕ diagonal, (**??**) reduces to the existing `gaussian_nll_loss`, applied to each component.

Proposition 1 (What the Gaussian head learns)**.** *Fix $\mathcal{D}$ and let $\boldsymbol{m} = \mathbb{E}[\boldsymbol{y} \mid \mathcal{D}]$ and $C = \mathrm{Cov}[\boldsymbol{y} \mid \mathcal{D}] \succ 0$. Over all $\boldsymbol{\mu} \in \mathbb{R}^2$ and $\Sigma \succ 0$,*

$$\arg\min_{\boldsymbol{\mu},\Sigma}\ \mathbb{E}_{\boldsymbol{y}|\mathcal{D}}\big[-\ln\mathcal{N}(\boldsymbol{y};\boldsymbol{\mu},\Sigma)\big] = (\boldsymbol{m},\ C), \tag{18}$$

whether or not $p(\boldsymbol{y} \mid \mathcal{D})$ is Gaussian.

Proof. Using $\mathbb{E}\big[(\boldsymbol{y}-\boldsymbol{\mu})(\boldsymbol{y}-\boldsymbol{\mu})^\top\big] = C + (\boldsymbol{m}-\boldsymbol{\mu})(\boldsymbol{m}-\boldsymbol{\mu})^\top$,

$$\mathbb{E}\big[-\ln\mathcal{N}\big] = \tfrac{1}{2}\operatorname{tr}\big(\Sigma^{-1}C\big) + \tfrac{1}{2}(\boldsymbol{m}-\boldsymbol{\mu})^\top\Sigma^{-1}(\boldsymbol{m}-\boldsymbol{\mu}) + \tfrac{1}{2}\ln\det\Sigma + \text{const.} \tag{19}$$

The second term is non-negative and vanishes only at $\boldsymbol{\mu} = \boldsymbol{m}$. With $P = \Sigma^{-1}$, the remainder $\frac{1}{2}\operatorname{tr}(PC) - \frac{1}{2}\ln\det P$ is strictly convex in P, and its gradient $\frac{1}{2}(C - P^{-1})$ vanishes only at $P = C^{-1}$, i.e. $\Sigma = C$. □

Consequently, with enough capacity and data, the optimum of (**??**) ($\lambda \to 0$) satisfies $\boldsymbol{\mu}_{\hat\phi}(\mathcal{D}) \approx \mathbb{E}_\pi[\boldsymbol{y} \mid \mathcal{D}]$ and $\Sigma_{\hat\phi}(\mathcal{D}) \approx \mathrm{Cov}_\pi[\boldsymbol{y} \mid \mathcal{D}]$. The head matches the posterior's first two moments. If the posterior is skewed or curved (the *a*–*b* degeneracy), Gaussian credible regions are miscalibrated even at the optimum.

4.2 Flow head (exact conditional density)

Let $f_\phi(\cdot\,;\boldsymbol{c}) : \mathbb{R}^2 \to \mathbb{R}^2$ be a diffeomorphism for every $\boldsymbol{c}$, and $\boldsymbol{z} = f_\phi^{-1}(\boldsymbol{y};\boldsymbol{c})$. Then

$$-\ln q_\phi(\boldsymbol{y} \mid \mathcal{D}) = \tfrac{1}{2}\|\boldsymbol{z}\|_2^2 + \ln(2\pi) - \ln\left|\det\frac{\partial f_\phi^{-1}(\boldsymbol{y};\boldsymbol{c})}{\partial \boldsymbol{y}}\right|. \tag{20}$$

If the flow family is rich enough to contain $p_\pi(\boldsymbol{y} \mid \mathcal{D})$, the minimizer of (**??**) is exact: $\mathcal{R}(\hat\phi) = 0 \iff q_{\hat\phi}(\cdot \mid \mathcal{D}) = p_\pi(\cdot \mid \mathcal{D})$ for $p(\mathcal{D})$-almost every $\mathcal{D}$.

5 Inference on a new event

For an observed $\mathcal{D}^*$ the estimate costs one forward pass:

$$\hat{\boldsymbol{y}} = \boldsymbol{\mu}_{\hat\phi}(\mathcal{D}^*), \qquad \hat\Sigma = \Sigma_{\hat\phi}(\mathcal{D}^*). \tag{21}$$

Derived quantities. Because $(\ln m,\ \ln\sigma) = B\boldsymbol{y}$ with $B = \begin{pmatrix} 1 & 1 \\ \frac{1}{2} & 1 \end{pmatrix}$, the Gaussian family is closed under this map:

$$(\ln m, \ln\sigma) \mid \mathcal{D}^* \ \approx\ \mathcal{N}\big(B\hat{\boldsymbol{y}},\ B\hat\Sigma B^\top\big), \qquad \mathrm{Var}(\ln m) = \hat\Sigma_{11} + \hat\Sigma_{22} + 2\hat\Sigma_{12}. \tag{22}$$

On the original scale $a = e^{y_1}$ is log-normal:

$$\mathbb{E}[a \mid \mathcal{D}^*] = \exp\big(\hat y_1 + \tfrac{1}{2}\hat\Sigma_{11}\big), \qquad \mathrm{median}(a \mid \mathcal{D}^*) = e^{\hat y_1}. \tag{23}$$

6 Validation

Calibration. If $q_{\hat{\phi}}$ is the true Gaussian posterior, the squared Mahalanobis distance of held-out truth satisfies

$$r_e^2 = (\boldsymbol{y}_e - \hat{\boldsymbol{\mu}}_e)^\top \hat{\Sigma}_e^{-1} (\boldsymbol{y}_e - \hat{\boldsymbol{\mu}}_e) \;\sim\; \chi^2_2, \qquad \Pr\left(r_e^2 \le \chi^2_{2,\alpha}\right) = \alpha \quad \forall\, \alpha \in (0,1). \tag{24}$$

Compare the empirical CDF of $\{r_e^2\}$ with χ^2_2. An excess at large r^2 means the posteriors are too narrow. For the flow head, check coverage of highest-density regions, or use simulation-based calibration ranks.

Resolution floor. By the law of total variance,

$$\mathrm{Var}(\boldsymbol{y}) = \underbrace{\mathbb{E}\big[\mathrm{Var}(\boldsymbol{y} \mid \mathcal{D})\big]}_{\text{irreducible given } \mathcal{D}} + \mathrm{Var}\big(\mathbb{E}[\boldsymbol{y} \mid \mathcal{D}]\big). \tag{25}$$

No estimator of $\boldsymbol{y}$ from $\mathcal{D}$ has mean squared error below $\mathbb{E}[\mathrm{tr}\ \mathrm{Var}(\boldsymbol{y} \mid \mathcal{D})]$. The shower-to-shower fluctuations that leave no trace in the light set this floor.

Label noise. The labels (**??**) are finite-sample fits, $\boldsymbol{y}_e = \boldsymbol{y}_e^{\mathrm{true}} + \boldsymbol{\epsilon}_e$. The network therefore learns $p(\boldsymbol{y}^{\mathrm{label}} \mid \mathcal{D})$, whose covariance includes $\mathrm{Cov}(\boldsymbol{\epsilon})$. This matters only if the fit error is comparable to the posterior width.